Velivo 1.22

User manual – every feature explained

Part I – User manual · Part II – Technical appendix (detailed description)

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Part I

User manual

Complete guide and user manual

Made with passion. Author: **Michael** (github.com/mic0078).

Polski | **English** · Project home

Velivo is a private browser for Windows that keeps your data with you. There is no account and no cloud, and your history and passwords are never sent to outside servers. Pages are displayed by the Microsoft Edge engine (WebView2), so they look and work exactly as in Edge and Chrome. Everything around the pages – the window, tabs, menus, privacy, downloads and sync – is Velivo's own code.

Velivo is designed to be fully usable with the mouse alone, even from the sofa in front of the TV. It works in Polish and English.

1. What makes Velivo different

Velivo goes beyond the limits of today's browsers. Chrome, Edge, Firefox and Opera **do not have these features built in**. Some need separate extensions, some can't be done at all. In Velivo they work right away, with no add-ons and no account.

Feature	What it does
All in one	Video player for files on disk (with picture in picture and "Video on top"), PDF viewer and torrents – **no separate programs to install**. Open files by double-click or "Open with".
▣ **Video on top**	Velivo's own video window you can shrink almost to an icon (from 160×90 px), with **transparency set by the mouse wheel** and an "always on top" pin. **It keeps playing after you close the tab – and even the whole browser.**
"Where did I read that?"	Finds an article you read by words from its **content**, not just its title. Everything stays on your computer.
⚠ **Shop-trick detector**	Warns about fake countdowns, "only 2 left" pressure and hidden fees, and unticks pre-selected add-ons in the cart.
Privacy receipt	Shows how many companies and countries the page connected to, which of them are data brokers and whether it tried to fingerprint your computer.
Send tabs at home	The page opens on another computer running Velivo, and videos continue at the same moment. No cloud, no account.
Sync without the cloud	Bookmarks, passwords, history, pinned tabs, Quick Access and settings between home computers, encrypted.
⬇ **IDM-style video downloads**	A "Download" button over every video, YouTube included, with quality, MP3/M4A and folder choice.
"Read from here"	Right-click a paragraph and Velivo reads from that sentence.
Night mode with strength	Like Windows "Night light", adjusted with the mouse wheel.
Offline fraud protection	Recognises fake bank, shop and portal sites (`paypa1.com`, `ebay-verify.top`, letters from other alphabets) and fakes of sites you have saved passwords for – before the page opens.
Safe payments	On bank and payment sites the Velivo window becomes invisible to screen-recording programs.
Fake-proof passwords	A password goes only to the real domain. A page cannot swap the helper, "click" for you or steal data with a hidden field.
Moving from another browser	Bookmarks read directly from Chrome, Edge, Brave, Opera and Vivaldi; passwords from KeePassXC and others via CSV.
Gestures and mouse-only use	Back, forward, new tab, close, reload and zoom without a keyboard.

2. How small it is and why

Item	Size
Installer `Velivo-Setup-1.22.exe`	**about 60 MB** – including the built-in .NET 10 and the uBlock Origin Lite rule lists; nothing else to install
Installed program	about 190 MB (including .NET 10 and the Quick Access add-on)
RAM used by Velivo itself (`Velivo.exe`)	typically **50–80 MB**
For comparison: Chrome or Firefox installer	about 100–130 MB

Why the installer is so small

1. **Velivo doesn't ship its own browser engine.** Chrome, Opera and Brave each include the whole Chromium engine, over 100 MB. Velivo uses the Microsoft Edge engine (WebView2), which is already part of Windows 10 and 11 and is updated by Windows Update. Velivo only adds its own parts: the window, tabs, privacy, downloads and sync.
2. **The interface is a native Windows program (WPF, .NET),** not a web page pretending to be an app as in Electron. Native code like this is very compact.
3. **Large extras download only when you need them, and only with your consent:**
   - full ad-block lists download in the background after the first start;
   - natural offline voices are about 60 MB each;
   - the yt-dlp video tool is about 18 MB;
   - the FFmpeg add-on for best quality and MP3 is about 140 MB.

Why Velivo uses so little memory

1. **The window, tabs, menus and toolbars are native.** They are not built from web pages, so the shell itself takes 50–80 MB instead of several hundred.
2. **Ads and trackers are blocked before they download.** Pages carry fewer scripts, images and frames, so they use less memory and load faster.
3. **Background tabs give memory back.** Velivo tells the engine a tab is in the background and the engine frees part of its RAM. Music and chats in those tabs keep working.
4. **There are no background services:** no account, cloud, telemetry, suggestions, shopping or news feed.
5. **The engine is shared with Windows.** Part of its libraries are kept in memory by the system anyway.

To be fair: the 50–80 MB is the Velivo program itself. Page content is rendered by the Edge engine in separate `msedgewebview2.exe` processes, which also use memory, as in every browser. How much depends on how many tabs are open and how heavy the pages are. Thanks to ad blocking and background-tab sleep it is usually less than Chrome uses for the same pages.

3. Installing and updating

Requirements:

- Windows 10 version 1809 or newer, or Windows 11, 64-bit;
- internet during setup only if WebView2 is missing or you choose the video player.

.NET 10 is built into Velivo – no need to install it. **Microsoft Edge WebView2** (the page engine) is in Windows 11 and current Windows 10; if it is missing, the installer downloads it from Microsoft and installs it. The VLC-like video player (about 145 MB, official VideoLAN package) is installed only if you tick it in the installer.

Step by step:

1. Download `Velivo-Setup-1.22.exe` from the `Instalator` folder.
2. Run it. If Windows shows "Unknown publisher", click **More info → Run anyway**. The installer has no paid code signature.
3. **Choose the installer language: Polski or English.** Velivo will start in that language. You can change it later in Settings → Look → Language.
4. If Velivo is already installed, choose:
   - **Keep my settings and data** – a normal update (recommended);
   - **Clean install** – start from scratch. Old data is backed up with a `.kopia-<date>` suffix.
5. Optionally tick the desktop shortcut and finish.

No administrator rights are needed. The program goes to `%LOCALAPPDATA%\Programs\Velivo`. The installer closes Velivo processes running in the background and works around files locked by antivirus software.

Make it the default browser: Settings → "Set as default". Links from other programs will then open in Velivo.

4. Getting started – the toolbar

Velivo starts maximized. From the left:

Item	What it does
← →	Back and forward
⟳	Reload (F5)
⌂	Home page
Address bar	Type an address or search words. Right-click gives Cut, Copy, Paste and **Paste and go**
125%	Page zoom. The **mouse wheel over the button** changes it and a click restores the default. Zoom is remembered per site
****	Appears when Sejf has logins for the site; a click fills them in
☆	Add or remove a bookmark (Ctrl+D). A gold star means the page is bookmarked
Privacy	Rules panel for the current domain
Extension icons	Chrome extensions. They can be pinned and unpinned
+ number	Shield: how much was blocked on the page. A click shows the list and the privacy receipt. A red shield means AdBlock is off
⚠ **+ number** (yellow)	Appears in a shop that uses pressure tricks
****	All bookmarks
☀ / /	Page mode – the icon shows the current one: ☀ light → dark → night. In night mode the mouse wheel sets the strength
****	Screenshot: visible part or the whole page
Reader	Reader mode with a summary
****	Read the page aloud. While reading, pause, stop and speed buttons appear
⬇	Downloads (Ctrl+J) with the number of active downloads
****	Extensions: list with pins and "Manage extensions..."
****	History (Ctrl+H)
⚙	Settings
Profile	Active profile; a click switches

The tab bar also has **+** (new tab, Ctrl+T) and **** (private tab, Ctrl+Shift+N).

On a narrow screen some buttons move into the **"..."** menu.

5. Right-click menus

On a page

Right-click anywhere on a page.

Item	When it appears	What it does
Search "..." / Go to ...	Text selected	Searches the selection or opens the selected address
Translate selection / Read selection aloud	Text selected	Translates or reads the selection
Translate page to Polish / English	Page in another language	Google Translate for the whole page
Block element (ad)...	Always	Point at an element that should disappear on every visit
Restore blocked elements	When something was blocked by hand	Undoes the blocks on this page
(Don't) reject cookie banners	Always	Exception from automatic rejection for this site
/ Phone / desktop version	Always	Switches and remembers for this site
Page mode: ☀ Light / Dark / Night	Always	Pick this page's mode directly (like the toolbar button)
▣ Video on top · ⬇ Download video · Picture in picture	On a video or on YouTube	See section 8
Reader mode and summary	Always	Clean article text
Read page aloud (Ctrl+Shift+U)	Always	Reads the main content
Read from here	Nothing selected	Reads from the sentence you clicked
Where did I read that?	Always	Search in the content of pages you read
Screenshot	Always	Visible part or the whole page
Velivo tools	Always	Open a file (video / PDF), this page in a private tab, this page in Banking Mode, privacy, media, downloads, history, extensions, network diagnostics, profiles, ⚙ Settings, picture in picture, tab search
Add to Quick Access	Always	Shortcut to a chosen group

On a tab

Right-click a tab.

- Reload, Duplicate tab
- Pin or unpin tab
- Mute tab or unmute
- Auto refresh: every 1, 5, 15 or 30 minutes
- Send to... (another computer running Velivo at home)
- Add to group, Tab sets, Search tabs
- Close tab, other tabs or tabs to the right (pinned tabs are skipped)
- Reopen closed tab (Ctrl+Shift+T)

In the Modern look menus show Windows 11 icons. In the Colorful look they show emoji.

6. Mouse-only use: gestures and the wheel

Mouse gestures

Hold the **right button**, move the mouse about 3 cm and release:

Move	Action
← left	Back
→ right	Forward
↑ up	New tab
↓ down	Close tab
↓ then →	Reload

A normal right-click without moving opens the menu as always. You can turn gestures off in Settings → Search and start.

A **pinned tab** is not closed by the ↓ gesture (just like it has no close button). The turn in ↓ then → counts from about 15 px, so "Reload" doesn't accidentally become "close tab".

Mouse wheel on buttons

Where	Action
Zoom button (e.g. 125%)	Zooms the page in and out, remembered per site
Page mode button () in night mode	Warmth strength 5–100%
"Video on top" window	Transparency 15–100%

Clicking links

- **A normal click** opens in the same tab (Back and Forward work). You can change this in settings.
- **The middle button** or **Ctrl+click** always opens a new tab.

⌨ Velivo on-screen keyboard

On by default (Settings → "⌨ Velivo on-screen keyboard").

- **Opening:** middle-click a text field, hold the wheel for about 1 s anywhere, click the keyboard icon by the clock, or use the **Velivo Klawiatura** shortcut in the Start menu (right-click → Pin to taskbar). Optional: right-click the tray icon → **Open automatically when you click a text field**. The **—** button hides it to the tray; hidden, it keeps correcting what you type on a physical keyboard.
- **Correct spelling** (✓ abc in the header): after Space it fixes Polish letters and typos (zolw → żółw), „nie” with verbs (niewiem → nie wiem), fixed phrases (napewno → na pewno), „ze/że” by context, polite capitals (ty → Ty). ⌫ **right after a correction undoes it.**
- **Punctuation:** commas before że, który, bo, ale, a…; a comma or period sticks to the word („kot ,” → „kot,”); a space is added after a comma; capital letter after a period.
- **Suggestions:** the strip above the keys shows completions (Space inserts the first one), the next word you usually type, and whole sentences you type often (click or **Ctrl+Space**).
- **Learning:** your words, phrases, sentences and corrections (delete a word right after typing it and retype it – Velivo remembers the fix). Encrypted with your Windows account, on this computer only.
- **My dictionary:** a window with your words – paste many at once or load a .txt file, search, sort, clean up (removes ordinary words and typos). Add a new word with or **Ctrl+Space** right after typing it.
- **Swipe typing**, **numeric keypad** with Select all, copy, paste, undo, redo, **Alt+Tab** (window switcher), shortcut bar: Notes, sticky notes, e-mail, volume, calculator, snipping, quick search,

shutdown.

- **Gamepad** (Xbox / XInput): D-pad picks a key, A presses, B ⌫, X space, Y ⇧, LT Polish letters, LB/RB cursor, Start ⏎; **right stick moves the mouse cursor** (RT faster, held L3 slower); with the keyboard hidden A – click, X – right-click, Y – double-click, Start – Enter; **R3** – window switcher; **Back** – hide / show. Right-click – settings (cursor on/off, sensitivity, RT, L3).
- **Passwords are safe:** nothing is corrected, suggested or learned in password fields.

Sticky notes

- The button on the keyboard: a new note (plain, to-do, shopping, call, important, idea, today) or show all.
- Docked above the taskbar side by side; click the bar to expand upward, undocks to the desktop, color, always on top, inserts a date, double-click a line to tick it.
- **Reminder:** when a note contains a date (19.10.2026, 19.10, 2026-10-19), its icon blinks and its bar pulses the day before and on the day until you open it.

7. Tabs

- **Pinning:** right-click a tab → Pin tab.
  - the tab moves to the front of the bar, with a pin and no close button;
  - it is "frozen" to its address: a link to another site opens in a new tab;
  - it comes back on every start and syncs with your other computers.
- **Tab groups:** right-click → Add to group → New group….
  - tabs in a group have a colored dot, with a colored label in front of them;
  - a click on the label collapses the group into one label with a tab count;
  - right-click the label: name, color, ungroup, close the whole group.
- **Tab sets:** right-click → Tab sets → Save open tabs as a set…. From the same menu you later open the whole set, replace it or delete it.
- **Mute:** a tab that plays sound shows . A click mutes it ().
- **Auto refresh:** right-click → Auto refresh. The tab then shows ⟳.
- **Tab search:** Ctrl+Shift+A, type a few letters, Enter switches.
- **Right-click "+" or an empty spot on the tab bar:** new tab, new private tab, reopen closed tab, **Tab sets** (open a saved one / save current), **Tab groups** (expand / collapse), tab search.
- ◫ **Split screen:** right-click a tab → "Show side by side (split screen)" – two tabs next to each other, drag the middle bar; clicking the other page makes it active. Leave with "Close split screen" or by closing one of the tabs.
- **Sleeping tabs** (Settings → Tabs, off by default): a tab you haven't viewed for a while goes to sleep – less memory and CPU; it wakes on click. The viewed, pinned or playing tab never sleeps.
- **Private tabs ():** save nothing, don't sync and don't go to history.
- **After a restart** your tabs, groups and pinned tabs come back. You can turn this off in settings.

8. Videos: Video on top, picture in picture, downloads

Hover over a video and three buttons appear in its top right corner: **⬇ Download · ▣ On top · Picture in picture**. This works on YouTube too.

▣ Video on top (Velivo only)

1. Click ▣ **On top**. The video in the tab pauses and continues in a small window from the same moment.
2. Using the window:
   - **move** it by the dark bar at the top;
   - **resize** it by an edge or corner, from 160×90 px;
   - **transparency:** mouse wheel over the window, 15–100%; the bar briefly shows "Visibility 60%";
   - **pin:** blue means always on top, white means a normal window that can go behind others;
   - **click the video** to pause or resume;
   - ↩ goes back to the page in Velivo from the same moment, even if the browser was closed;
   - ✕ closes the window.
3. The window **keeps playing after you close the tab, and even the whole of Velivo.** Clicking the Velivo icon then starts the browser again.
4. Position, size, transparency and pin are remembered.

Clean video view. The window shows **only the video**, filling the whole window – none of the rest of the page: no menus, comments, suggestions, pop-ups or banners. On YouTube it feels like a separate player. Ad and tracker blocking works as in a tab, and YouTube video ads are skipped (clicking "Skip" or fast-forwarding the muted ad).

What it's great for:

- music or a podcast in a small, see-through window in a corner while you work in other programs;
- a match or live stream over a document or spreadsheet – set visibility to 40–60% and the text underneath stays readable;
- a video tutorial next to the program you're following it in;
- the video keeps playing even after Velivo is closed, so the browser uses no memory.

Mouse-only summary:

What	How
Pause / resume	click the video, ▶/❚❚ on the bar or Space
Seeking	◀◀ / ▶▶ (10 s), time bar, ← → keys, wheel over the time bar (5 s)
Volume	mute, slider, ↑ ↓ keys, wheel over the slider, M
Full screen	on the bar, F or double-click the video; Esc to leave
Transparency 15–100%	mouse wheel over the window
Move	dark bar at the top
Size (from 160×90)	edge or corner
Always on top	(blue = on)
Back to the page at the same moment	↩
Close	✕

Transparency needs Windows 10 version 1809 or newer, because the window uses a Windows component to draw the picture.

Video player and TV

- **Opening:** double-clicking a video file in Windows opens just the Velivo player – no browser window. Also **Ctrl+O** or right-click → Velivo tools → **Open a file**.
- **VLC-like player** (AVI, MKV, WMV, MP4 and more) – optional in the installer or later in Velivo (about 145 MB from the official VideoLAN package). Without it, MP4 and WebM play in a browser tab.
- **Video library**, **TV** with guide, catch-up and recording, **picture in picture**, sleep timer and scheduling, videos from another computer at home.
- **Subtitles:** right-click the video or the library → **Download subtitles…** (OpenSubtitles.com, sign in with your own account; the password is encrypted on this computer). Subtitles are saved next to the film and load automatically. **G / H** shifts subtitles – remembered per film.
- **Narrator** reads subtitles aloud (Piper offline voice or a Windows voice); settings remembered per film.
- **If Velivo isn't in "Open with":** start Velivo once or use **Settings → Video player → "Open videos and PDFs in Velivo (Windows)…"**.

Picture in picture

The standard window of the browser engine. It keeps playing after you close the tab, until you close the window. Its minimum and maximum size are set by the Chromium engine, not by Velivo.

⬇ Video downloads

1. Click **⬇ Download** or right-click a video → **⬇ Download video…**.
2. Choose the quality:
   - **best video quality** (up to 1080p, MP4);
   - **quick** – video in one file, usually 360p–720p;
   - **audio only M4A**;
   - **audio only MP3**.
3. Choose the folder (**Change folder…**). Velivo remembers the last one.
4. Progress, speed and time left are shown in the Downloads window.

What does the downloading:

- **Regular video files:** Velivo's manager, up to 16 connections at once.
- **YouTube, streams and over 1,000 sites:** the free **yt-dlp** tool:
  - downloaded once with your consent (about 18 MB);
  - updates itself every 2 weeks;
  - downloads 8 parts at once.
- **Best quality and MP3** need the free **FFmpeg** (one-time, about 140 MB, with consent).

Limits:

- Netflix, Disney+ and other DRM-protected services are not supported.
- Only download content you have the right to, for example for personal use.

9. Privacy and security

Feature	Description
Ad and tracker blocking	EasyList, EasyPrivacy and a Polish list, about 95k rules. Works before download, so pages are lighter
Manual element blocking	Right-click → Block element. The mouse wheel widens the area, a click blocks, Esc cancels
Cookie banners (GDPR)	Velivo clicks "Reject" or "Only necessary" for you. **It never clicks "Accept".** If a banner has no reject button, nothing is clicked
Tracking prevention	Balanced (default) or strict. On trusted sites strict behaves like balanced, unless you tick "Force tracker blocking"
Site rules (Privacy button)	Block JavaScript or cookies, force tracker blocking, auto-clear data, trusted domain
Shield	Counter and list of everything blocked: AdBlock, rules, JavaScript, hidden elements, cookie banners
Privacy receipt	At the top of the shield window: how many outside companies, in how many countries, data brokers, fingerprinting attempts (canvas, graphics card, audio) and the most contacted companies
"Do Not Track"	Sends DNT and Global Privacy Control signals
SmartScreen	Microsoft protection against dangerous sites and files, skipped on trusted domains
Fake-site detection (offline)	Before a page opens, checks whether the address imitates a bank, shop or portal: a brand on a foreign domain (`paypal-secure-login.com`, `ebay.co.uk.account.top`), typos and swapped characters (`paypa1.com`, `rnicrosoft.com`, `amaz0n.com`), `xn--` addresses with letters from other alphabets, and fakes of sites you have saved passwords for. Shows a warning with "No" as the default. If it's the real site, choose "Yes" – Velivo remembers it
Always HTTPS	Every connection is tried encrypted first. An unencrypted site opens only after a warning, so you don't type passwords or card details there
Safe payments	On bank and payment sites (PayPal, Monzo, Revolut, Barclays, HSBC, Lloyds, NatWest, Santander, PKO, mBank, Stripe and more) the Velivo window is invisible to screen-recording programs. A message appears. You can't take screenshots on those sites
Attack-proof passwords and forms	Passwords only for the real domain; the Velivo helper can't be swapped by a page; only real mouse clicks work; hidden fields are never filled; Velivo asks before filling a card or bank account; the CVC is never saved
Private tabs	Separate, isolated data that disappears when closed. Passwords and forms work; saving only with your consent
Clearing data	On exit or by hand in settings

All switches are in Settings → "Security and downloads". Attack tests are in the repository under `testy/bezpieczenstwo/`.

Honestly: Velivo protects you in the browser. Viruses and spyware on the system itself are caught by an antivirus (e.g. Windows Defender or Bitdefender) – keep it on.

10. Shopping: pressure-trick detector

When a shop tries to rush you, a yellow ⚠ **number** button lights up next to the shield. A click shows the list:

- **a countdown.** If it restarts after you reload the page, Velivo marks it as **fake**;
- **"only 2 left", "3 seats remaining";**
- **"15 people are viewing now", "someone just bought";**
- ☑ **pre-ticked insurance, warranty or newsletter in the cart.** Velivo **unticks** them;
- **a service, handling or booking fee** that appears only at checkout.

The detector reads the page's texts and behavior, so it won't catch every shop and can occasionally be wrong.

11. Reading: read aloud, reader mode, "Where did I read that?"

Read aloud

- **on the toolbar** or Ctrl+Shift+U reads the main content. The paragraph being read is highlighted and scrolled to the middle of the screen.
- **Right-click → Read from here** reads from the clicked sentence. A click elsewhere jumps there.
- **Select text → Read selection aloud** reads just that part.
- **Same content as Reader:** read aloud and Reader share one article detector – only the article is read, without side columns and extras.
- **Voice per paragraph language:** in mixed text each paragraph is read with a voice in its own language (Polish / English).
- **Reacts to changes:** a new voice or speed in Settings applies at once; when the page switches to another article, reading stops.
- While reading, the toolbar shows pause, stop and speed (0.75×–2×); volume is in settings.
- **Voices:**
  - Windows Polish and English voices;
  - natural Piper offline voices: Gosia, Darkman, MC Speech, Lessac, Ryan, Alba. Each downloads once, about 60 MB.
- **Skipped while reading:**
  - menus, footers and ads;
  - "Read also" and "Recommended" blocks and link lists;
  - comments, newsletters and photo captions.

Reader mode ("Reader" button)

- Clean article text in a readable font, without ads or distractions.
- **"In short"** – a few key sentences of the article, computed on your computer. Only complete, self-contained sentences: no subheadings, torn-out quotes or sentences that make no sense without the previous one. If there are none, the box is simply not shown.
- Reader finds the article even when the page hides it in an unusual block (e.g. next to X posts or videos) – instead of "not enough content" it shows the text without menus and recommendations.
- "Read summary" and "Read all" buttons. Clicking the text reads from that point.

"Where did I read that?" (Ctrl+Shift+F)

- Velivo remembers the text of pages you read, **only on this computer**.
- Type words you remember from the content, for example "laptop battery 6 hours". Letter case and Polish characters don't matter.
- Results show the title, site, date and a snippet with the words. A double-click opens the page.
- **Skipped:** private tabs, banks, payments, mail, logins, gov.pl, ZUS and any page with a password field.
- Only the page's **main content** is stored (no menus or ads); **search result pages** are not stored.
- **Where to open it:** right-click on a page, the button in the History window or Velivo tools.
- Clearing history clears this memory too. You can turn the feature off in settings.

12. Search and the address bar

- **Type words** instead of an address and Velivo searches them in your chosen search engine (Settings → Search and start).
- **Search shortcuts:** type a shortcut, a space and the search text:

Type	Searches
`yt cats`	YouTube
`g recipe`	Google
`ddg weather`	DuckDuckGo
`wiki Kraków`	Wikipedia (PL)
`allegro bike`	Allegro
`olx sofa`	OLX
`ceneo tv`	Ceneo
`mapy Gdańsk`	Google Maps
`filmweb Matrix`	Filmweb
`tlumacz hello`	Google Translate

Add your own in Settings → Search and start → **Search shortcuts…**, for example `shortcut=address with %s`.

- **Right-click the address bar → Paste and go** pastes and opens at once.

13. Look: styles, themes, dark and night mode, zoom

- **Style** (Settings → Look):
  - **Modern** (default): calm buttons without colored backgrounds, Windows 11 icons, one blue accent, and color only on hover or when something is on;
  - **Colorful**: colored buttons and emoji.
- **10 themes:** Light, Graphite, Navy, Night violet, Forest, Ocean, Sunset, Black (OLED), Paper, Mist.
- **Page modes** (mode button – the icon shows the current mode: ☀ light, dark, night):
  - light;
  - **dark** – pages darkened while photos keep their true colors;
  - **night** – warm colors with less blue light, **strength set with the mouse wheel**.
- **Each site has its own mode** – one light, another dark, another night; Velivo remembers it per site. In dark mode a page enters without a white flash (dark background before painting, dark mist in the entrance effect).
- **Zoom** remembered per site, changed with the wheel on the percent button or with Ctrl+wheel.
- **Phone version** for a chosen site (right-click), remembered.
- **Language:** Polish or English (Settings → Look → Language).
- **Content entrance effect:** a new page appears from a focus-in blur, from darkness (cinematic), with a gentle dim or instantly; speed 0 to 5 s (Settings → Look). Details in chapter 21.
- **Reader mode** has its own look: ☀ light, dark or night with a strength slider.

14. Quick Access – the new tab page

- Shortcuts arranged in groups, for example Start, Finance, Music, Games.
- Site icons and **page thumbnails** made in the background. Icons don't flicker when you switch groups.
- **Quick Access profiles**, optionally with a PIN, and **background themes**.
- Adding: the **+ Shortcut** button, right-click on a page → "Add to Quick Access", or import bookmarks.
- Syncs with other home computers: shortcuts and groups are merged.

15. Passwords and Sejf

- **Sejf** (a separate program by the same author) keeps passwords encrypted offline. When a page has a login field, **** appears on the toolbar and a click fills in the form.
- **Saving to Sejf:** after you log in, Velivo asks whether to save or update the password.
- **Velivo's own encrypted store** (Local password manager) with a **strong password generator**.
- **CSV import and export**, for example from KeePassXC or Chrome.
- **Import from Chrome/Edge/Brave/Opera/Vivaldi** (Settings → “Import from Chrome/Edge/Brave…”): Velivo reads bookmarks by itself, from every profile. Passwords move via a CSV file: a button opens the password export page in that browser, then “Load CSV file…”, and finally delete the CSV file.
- **Important:** import **before** you uninstall the old browser. Once it is removed, its bookmarks can't be read and its passwords can't be exported. Passwords from a Google account can also be downloaded later from passwords.google.com. Site logins don't transfer – sign in once in Velivo.
- **Buttons by the login field:** a **Fill in · ⚡ Generate** badge appears to the right of the field:
  - **Fill in** lists this site's accounts (name and e-mail); a click fills in login and password. At the end of the list, **Another account from the Velivo vault…** opens a window that searches the whole vault (e.g. signing in with a PreSonus account on Fender's site). The window shows the site address and warns about fakes;
  - ⚡ **Generate** types a strong random password.
- **Save prompt:** after you click "Sign in", press Enter or submit a form, Velivo asks whether to save or update the password. Works in private tabs too.
- **Password manager:** search, copy login and password, edit, export. Entries from KeePassXC and phones without an address get a domain from the title or app link (`android://…`), and a double-click opens the site and fills the form.
- **Autofill** (local, encrypted store): **address** (first and last name, street, postcode, city, phone, e-mail), **card** (no CVC) and **bank account** (IBAN, account number, sort code). Clicking an empty field fills the whole form – on any https site. Card and bank details are filled only after you confirm the site address.

16. Banking mode

Banking mode is a separate, isolated browser profile for banking, payments and shopping, with its own **encrypted vault**. Normal tabs can't see anything from it and vice versa. It has no extensions and no history, and the cache is cleared when it closes (logins and "remember me" stay).

Opening and settings

- The green ** **button on the tab bar. The first time you set a** password **(min. 8 characters) and, optionally, a** hardware key **(YubiKey, Google Titan and other FIDO2 keys): tick "Also require a hardware security key", insert the key and click** " Add key" **– Windows asks you to touch it twice. Add all your keys in the same window, then** "Save"**.
- **Key plugged in** – Velivo asks you to touch it right away, no password needed. **No key** – enter the password. A key marked **** opens the mode on its own and encrypts the vault.
- Banking tabs are green and show . The mode **locks itself** after inactivity (1–60 min, your choice) and closes all its windows.
- Everything is under **right-click on** , including **Banking mode guide**.

My banks and My online shops

- **My banks / My online shops** – your sites; a click opens the site in banking mode.
- Adding: **" Add this site to My banks / shops"** on an open banking tab, or **" Add a bank / shop (name and address)…"** in the list.
- ✏ **Login details** (in My banks): login / customer number, passcode / PIN, password and memorable information – separately for each bank. Always enter the **full** password and passcode.
- **Fill in a login** – fills the login and password on the bank's page.
- **Fill in selected characters** – when the bank asks e.g. for the 2nd, 5th and 9th character (RBS, NatWest, Bank of Scotland, TSB, Lloyds, Halifax), Velivo reads the numbers and types the right characters, drop-down lists included. If a page doesn't work, **Copy a description of the login form** copies only the field description (without your data) for a report.
- **Data linked to the page** – on a page from your vault the login, password and selected characters fill in by themselves (when one account matches; Velivo never submits the form). Several accounts (e.g. personal and business) – one click to choose; payment card – always after a click. Can be turned off in Bank mode settings.
- **Full isolation** – bank mode uses only its own encrypted vault: the browser's regular passwords and autofill are neither suggested nor saved there, and vault data goes only into bank tabs (no extensions).
- **Extensions** get no access to bank or private tabs – even when opened from the toolbar while such a tab is active.
- **Transfer guard** – a pasted or typed account number is checked (IBAN / NRB checksum) and compared with your Bank accounts: a typo, a number swapped in the clipboard by malware or an unknown payee shows up at once.
- **Transfer from the vault** – on a transfer form Velivo fills in the payee, account number (NRB / IBAN / sort code + 8 digits) and reference, and the amount and customer number from the Bill with the same name. You always confirm it yourself.
- **Suggestion under the field** – click the login, password, selected characters, card number or payee account field: if the vault has matching data for this page, a choice appears right under the field (one click fills it); nothing matches – nothing appears. The page can't see your account names.
- **Menu only with matching data** – fill items in the menu appear only when the vault has data for the open page.
- **Safe filling** – the full password never goes into one-character boxes (only the selected characters do), and card data goes only into visible fields – hidden fields on the page get nothing.

Cards, notes and search

- **My cards** – click a card to view and edit it; **" Show number and CVV"** reveals the hidden fields. The CVV is optional. **Copy number / expiry / CVV** – the clipboard clears itself after 30 s. **Fill in a card on this page** fills the payment form. When the mode opens, Velivo reminds you of cards that expire soon.
- **My notes** – logins, passwords, customer numbers, with categories (Login, PIN, Transfers, Recovery codes, Other). **Generate password** inserts a strong password, **Numbered characters** shows the password character by character with numbers. The notes and search windows don't block the page.
- **Search my vault** – one field for banks, shops, cards and notes.

My logins and passwords

- For any non-bank service (email, shops, Netflix, government sites…): Name, Login, Password (hidden – " Show hidden data" reveals it), Website address, Note. **** next to each field copies it (clipboard cleared after 30 s).
- **Open site** – opens the address in a tab. **Fill in on the open page** – types login and password, but **only on a page with the same address (domain)** – a password never reaches a fake site. **Generate password** – inserts and copies a strong 20-character password.

Confidential data (documents)

- Document type (ID card, passport, driving licence, PESEL, NI number, health insurance, other), Full name, Number (hidden), Valid until (DD.MM.YYYY), Note.
- When banking mode opens, Velivo **reminds you about documents expiring within 60 days**.

Bank accounts

- Your accounts and payees (e.g. your landlord): Name, Account holder, Account number / IBAN (hidden), Sort code / BIC (SWIFT), Bank, Payment reference / note. **** buttons let you copy details into a transfer without mistakes.

Bills to pay

A list of regular and one-off payments (rent, electricity, heating, council tax, internet, instalments…), encrypted with the rest of the vault. Open it from **→ Bills to pay**.

- **Bill fields:** What for (name), Amount, Due date (DD.MM.YYYY), Repeat (weekly, every 2 weeks, every 4 weeks, monthly, every 2 months, quarterly, every half year, yearly, one-off – or type your own, e.g. "every 10 days"), Remind days before, Customer number / reference, Payment website, Note. Each field has **** to copy (clipboard cleared after 30 s).
- **Name from the sheet:** the What for field has a drop-down with the column names from the bills sheet (e.g. "Housing Rata 1", "Council Rata 2"). Pick the name from the list so paid bills land in the right sheet column. A different name (even a typo) = a new column in the sheet.
- **Add as new / ✓ Change selected / Delete selected / Clear fields** – normal list editing. Finally click **Save** – only then are changes (including payments marked as paid) written to the encrypted vault.
- **✓ Paid (next due date)** – select a bill and click: the payment is recorded in the history (month = due-date month, amount from the bill) and the due date moves on by the repeat interval. A one-off bill disappears from the list once paid. Clicking twice in the same month adds the amount twice – fix mistakes in the sheet.
- **Add bills from the sheet** – adds every sheet column that is not on the list yet: amount from the latest month, repeat monthly, due on the 1st of the following month. **Then correct the due day** and click **Save**.
- **Open payment site** – opens the Payment website address in a banking tab.
- **Reminders:** when banking mode opens, Velivo shows bills due within a few days and overdue ones – every day until you mark them as paid.
- Velivo **does not connect to your bank** – it cannot know you paid. You mark payments with Paid.

Bills sheet

An Excel-like overview of all paid bills, opened in a banking tab (local, offline). Button **Bills sheet (in a tab)** in the Bills to pay window.

- **How to read it:** each **row** is a month, each **column** is a bill, a cell holds the amount paid that month. The **Total** column on the right = month total, the **Total** row at the bottom = total of each bill across all months, bottom-right = grand total.
- **Typing:** click a cell and type the amount – totals update at once. Months can be written as words or numbers: "October 2026", "październik 2026", "10.2026", "2026-10".
- **Row (month)** – adds the month after the latest one in the table (after September 2026 → October 2026) and copies amounts from the last row (fixed bills need not be retyped).
- **Column (bill)** – a new bill; click the header to rename it. × under the name deletes the column, × at the end of a row deletes the month.
- **Load from Excel (CSV)** – in Excel: File → Save as → CSV (comma/semicolon delimited). Layout: **first column = month**, **first row = bill names**, amounts inside. The separator (`;`, `,` or tab) is detected automatically. Columns and rows named Total / Sum / Razem / Suma are skipped (so totals are not counted twice). Months already in the sheet are filled in (empty cells in the file do not erase your amounts), new bills become new columns and rows are sorted chronologically. The number of loaded rows is shown afterwards.
- **Currency you type amounts in** – £, zł, €, $, CHF, kr, Kč, Ft, lei, ₴, ¥. It is only a **symbol** – amounts are **not converted** by exchange rate. Default comes from Windows regional settings; the choice is remembered on Save and close.
- **Save and close** – the data is encrypted in the vault and the tab closes. Rows without amounts are skipped, as are rows without a valid month (Velivo tells you how many). **Cancel** closes without saving.
- **Link with Bills to pay:** the sheet and the Paid button share the same history. Every Paid shows up in the sheet in the due-date month, in the column with the same name.
- The older **Payment summary** window shows the same data as a table, with manual adding/removing of payments and **Save to Excel (CSV)** (export).

Protection

- **Fake sites:** while banking mode is open, a site that looks like your bank (e.g. another address with the bank's name) shows a big warning.
- **Bank in a normal tab:** when you open a listed bank, shop or payment site in a normal tab, a prompt above the taskbar asks **"Switch to banking mode"** / **"Stay here"**. Recognition works by the main domain, including other addresses of the same company.
- **Access log** – when, on which computer and how the mode was opened (wrong passwords included).

Profiles, backup and sync

- **Banking profiles** – a separate mode for another person: own password or key, separate logins and data.
- **Back up** and **Restore** – a `.vbank` file with all profiles, still encrypted (e.g. on a USB stick).
- **Sync** – banks, shops, cards, notes and mode settings go to paired computers in an encrypted package. Bank logins (cookies) stay on each computer.
- **Forgotten password:** without the password and the key the data can't be read. **"Forgot password – clear banking mode"** removes the mode and its data on this computer – so keep a second key and a backup.

17. Home network sync (no cloud)

Pairing (once)

1. Install the same Velivo version on both computers.
2. Use **the same profile** on both (Settings → User profiles).
3. Tick "Enable sync between running Velivo". The Windows firewall must allow Velivo (port 41919).
4. Velivo asks: "Connect both computers…?" Choose **Yes**, or click "Pair a device on the network…".
5. Compare the short code on both screens and confirm.
6. Choose whose settings to keep.

From then on everything goes **encrypted, over your network, without the internet**.

What syncs

Data	How
Bookmarks	Merged from both; deleting works on both
Velivo passwords	Merged; for the same account the newer one wins
History (last 60 days)	Merged; deleting an entry, day or everything works on both
Pinned tabs	The latest change wins
Quick Access	Shortcuts and groups merged
Settings, privacy, profiles, extensions	The latest change wins; a fresh install never overwrites data
Language, folders, video window	**No** – each computer keeps its own
Other open tabs	**No** – instead right-click a tab → Send to…

Sending tabs

Right-click a tab → **Send to… → computer name**. On the other computer the page opens in a new tab and the window comes to the front. Videos continue at the same moment.

The connection status is in Settings → **LAN diagnostics panel…**.

18. History and downloads

History (Ctrl+H)

- Arranged **day → site → pages**, for example "Yesterday · 180 pages · 12 sites" and inside it "polsatnews.pl · 45 pages".
- Repeated visits are shown once with a ×3 note. Quick Access entries are hidden.
- **Right-click:** open, open in a new tab, open all pages of a site, delete an entry, site or day.
- Search by titles and addresses, plus the **Search page contents…** button.

Downloads (Ctrl+J)

- **Velivo's manager:** up to 16 connections, pause and **resume after restart**, retry after a dropped connection.
- **Download history survives restarts.** Each file shows its size, date and time, and the site it came from.
- A deleted file is marked in gray: "File deleted or moved".
- Buttons: Open, Show in folder, Remove from list, Clear finished, Media on page.

Torrents (optional)

Turn them on in **Settings → Torrents** (off by default). Then:

- **magnet links** (clicked or pasted into the address bar) and **downloaded .torrent files** are downloaded by the free **aria2** – fetched on first use with your consent (about 2.5 MB) and run only if its SHA-256 checksum matches;
- **Windows association:** the installer associates `.torrent` files and `magnet:` links with Velivo – double-clicking a file or a magnet link from another program starts the download (if torrents are off, Velivo tells you where to turn them on);
- **separate zone:** files go to their own folder (default `Downloads\Velivo-Torrenty`); with "Ask where to save" on you pick a folder for each torrent; files are marked as from the internet and Velivo opens nothing by itself;
- settings: zone folder, download and upload speed, seeding after download (ratio and time), number of peers;
- buttons in Downloads: Stop; while seeding – Open folder and Stop sharing; at the end – Show in folder (selects the file), Delete file (removes the downloaded file or torrent folder from disk after confirmation) and Remove from list;
- after a stop or an error Velivo asks whether to delete the partly downloaded files;
- **a page can't start a torrent by itself:** a magnet link without your click or a pushed `.torrent` file – Velivo asks first; torrents never start from Banking Mode; aria2 reads no foreign config and keeps its data only in Velivo's folder;
- when Velivo stays in the tray, torrents keep downloading after the window is closed; "Exit completely" stops them; torrents don't work in Banking Mode.

Note: in torrents other peers can see your IP address – use a VPN for anonymity. Share only material you have the rights to.

19. Extensions, profiles and tools

- **Chrome extensions:** install from the Chrome Web Store, from a link or from an unpacked folder.
  - the button lists extensions with pins: pinned ones have an icon on the toolbar, unpinned ones keep working without the icon;
  - right-click an extension icon → Unpin from toolbar.
- **User profiles:** separate data per person or purpose, for example work or personal.
- **Screenshots:** the visible part or the whole page with scrolling.
- **Translation:** the whole page or a selection (Google Translate).
- **Detect media to download** on a page (Velivo tools).

uBlock Origin Lite (built in)

- Included in the installer and **enabled automatically** – no need to add it.
- Settings → "Security and downloads": a switch and a **"uBlock Origin Lite settings…"** button (filtering mode, lists, site exceptions).
- What it blocks adds to the **shield counter** and shows on its list. The shield tooltip shows "of which uBlock Origin Lite: N".
- **Updates itself** weekly from the developers' official GitHub. When the version is replaced, uBOL settings return to defaults.
- GPL-3.0 licence, details in `THIRD-PARTY.md`. If uBOL is already added from the store, Velivo keeps it (no duplicates).
- Add-on pop-ups fit their content, like in Chrome.

Cache on a RAM disk

If you have a RAM disk (for example ImDisk, SoftPerfect RAM Disk or your own tool), Velivo can keep its page cache there. **You don't need any external tools**, symlinks or moved folders, as you would with Chrome. Just pick the folder in Velivo's settings.

1. Settings → **Junk folder (empty = in the browser profile)** → **Choose…** and pick a folder on the RAM disk, for example `R:\`.
2. Save the settings and restart Velivo.
3. Velivo creates a `Velivo-smieci` subfolder there and when cleaning up deletes **only its contents**, never anything else on the RAM disk.
4. The "Delete junk on every browser start" option clears it on every start.

What you gain:

- cached pages load from RAM, which is faster than from disk;
- your SSD gets fewer writes;
- the cache disappears by itself when you turn the computer off, which is a bonus for privacy.

What it doesn't cover: the graphics cache (`GPUCache`) is always kept by the Edge engine in the browser profile. This is an engine requirement, not Velivo's choice. The **Default** button in settings restores the normal location.

20. Keyboard shortcuts

Shortcut	Action
Ctrl+T	New tab
Ctrl+Shift+N	New private tab
Ctrl+W	Close tab
Ctrl+Shift+T	Reopen closed tab
Ctrl+Tab / Ctrl+Shift+Tab	Next / previous tab
Ctrl+Shift+A	Search tabs
Ctrl+Shift+F	Where did I read that?
Ctrl+O	Open a file – video (player) or PDF
Ctrl+L	Address bar
Ctrl+D	Bookmark
Ctrl+H	History
Ctrl+J	Downloads
Ctrl+Shift+U	Read page aloud / pause
F5	Reload
F11	Full screen
Alt+← / Alt+→	Back / forward
Ctrl + mouse wheel	Zoom

You can also do each of these with the mouse alone: a button, a gesture or the right-click menu.

21. Settings – every option explained

Open the settings with the ⚙ button on the toolbar. Below is **every option in order**, as in the window.

Default browser

- Status: "✓ Velivo is the default browser" or "Velivo is not the default browser right now".
- **Set Velivo as the default browser…** – registers Velivo with Windows and opens the default-apps window.
- **On first start** (also right after installing) Velivo asks once whether to become the default browser; "No" – it won't ask again.
- **PDF:** Velivo opens PDF files in its built-in viewer (zoom, search, print, save). The installer adds Velivo to "Open with" for `.pdf`; make it permanent in Default apps → Velivo.

Import passwords, logins and bookmarks

- **Passwords and logins from a file** (CSV from KeePassXC, Chrome, Edge and other programs).
- **From Chrome / Edge / Brave / Opera** – bookmarks and passwords straight from an installed browser.

Look and readability

- **Default page zoom** – for all pages; each page can also be zoomed separately (Ctrl + wheel) and Velivo remembers it.
- **Dark page mode** – pages with their own dark look switch to it, other pages are dimmed (photos keep their real colours).
- **Night mode** – warmer colours and less blue light (like Windows Night light). The page mode button (☀ / /) cycles light → dark → night; change the strength with the mouse wheel on the button.
- **Content entrance effect** – how a new page appears on screen:
  - **Focus in** (default) – content emerges from a slight blur;
  - **From darkness (cinematic)** – the page brightens from a dark screen, like in a cinema;
  - **Gentle dim** – a short, subtle dim and back;
  - **None** – the page appears instantly.
- **Entrance effect speed** – slider from 0 to 5 s in 0.1 s steps (0 = automatic). Example values: quick 0.15 s, gentle 0.3 s, slow 0.5 s, calm 1 s, very calm 2 s, dreamy 3 s, slowest 4 s.
- **Browser theme** – colours of bars and tabs: Light, Graphite, Navy, Night violet, Forest, Ocean, Sunset, Black (OLED), Paper, Mist.
- **Style** – **Modern** (calm, like Windows 11: Windows icons, one blue accent) or **Colourful** (coloured buttons and emoji).
- **Interface language** – automatic (installer / Windows), Polish or English. Takes effect after a restart.
- **Always compact toolbar** – smaller labels, some buttons move into the "…" menu even on a wide window.

Read aloud

- **Voice and speed** – automatic (by page language: Polish/English), Windows voices, natural online voices or **natural offline voices** (Piper – downloads about 60 MB, then works without internet).

Tabs

- **Restore tabs from the previous session on start** – private tabs are never saved. Ctrl+Shift+T also restores a closed tab.

- **Open links in the same tab** – links a page wants to open in a new tab open in the current one (Back and Forward work). Ctrl+click still opens a new tab.

Search and start

- **Search shortcuts** – e.g. "yt cats" searches YouTube; add your own in the window.
- **Search engine in the address bar**.
- **Mouse gestures** – hold the right button and move: ← back, → forward, ↑ new tab, ↓ close tab, ↓→ reload. A normal right-click still opens the menu.
- **"Picture in picture" button over videos** – hovering a video shows ; the video moves to a small always-on-top window.
- **"Download" button over videos** – like Internet Download Manager: ⬇ Download over the video. Normal files go to the Velivo download manager (up to 16 connections), YouTube and streams to the free yt-dlp tool.
- **Start page** and **Quick Access as the new tab page** (separate Velivo data; Sejf's Quick Access in other browsers stays separate).
- **Extension folder for manual installation in other browsers** – a path to paste into "Load unpacked".

Privacy

- **Send "Do Not Track" signals** (DNT and Global Privacy Control).
- **Tracking protection** – **balanced** (recommended: blocks known trackers, embedded content like X posts or videos works) or **strict** (also blocks embedded social content; on trusted domains it acts as balanced unless you tick "Force tracker blocking" for the domain) or **off – no tracking control** (the engine blocks no trackers; uBlock Origin Lite and site rules still work if enabled).
- **Save browsing history**.
- **Clear data on close** – history and cache; accounts stay logged in.
- **Logins from Sejf: key on the toolbar on login pages** – clicking the key fills the form with a Sejf login.
- **Offer to save passwords**.
- **Form autofill** – addresses and cards in a local, encrypted offline vault. **Show saved data…**, **Delete saved cards**, **Delete saved addresses**.
- **Local password manager…**, **Import passwords CSV…**, **Export passwords CSV…**.
- **Block pop-ups opened without a click**.
- **Automatically reject cookie banners (GDPR)** – Velivo clicks "Reject" / "Necessary only"; it never clicks "Accept". Exception for a site: right-click on the page.
- **Remember the content of pages you read** – the "Where did I read that?" search (Ctrl+Shift+F), only on this computer.
- **Warn about pressure tricks in shops** – fake timers, "last items", "X people are viewing", pre-ticked extras (Velivo unticks them), hidden fees.
- **Privacy receipt on the shield** – how many companies and countries the page contacted, data brokers, attempts to fingerprint the computer.

Ad blocking

- **Full filter lists** (EasyList, EasyPrivacy, Polish list – about 97k rules). Downloaded in the background and refreshed every 4 days; **Update lists now** fetches them at once.

Security and downloads

- **Warn about dangerous sites and files (SmartScreen)** – addresses are checked with Microsoft.
- **Ask where to save each download**.

- **Detect fake bank, shop and portal sites** – works offline (paypa1.com, ebay-verification.top, fakes of sites you have saved passwords for).
- **Always encrypted connection (HTTPS)** – warns about unencrypted sites.
- **Safe payments** – on bank and payment sites the window is invisible to screen-recording programs (no screenshots there either).
- **uBlock Origin Lite** – built-in ad blocker (recommended, can be turned off), **uBlock Origin Lite settings…**.
- **Velivo speakers** – choose a specific sound output instead of the Windows default. **Don't lose sound** – when a music program (Ableton, Cubase) takes the speakers, Velivo plays on another active output and returns when they are free. **Windows volume mixer…** – pin Velivo to your speakers permanently.
- **Stay in the system tray when the window is closed** – background sync and instant start; the icon by the clock pulses while syncing, right-click: Open, Sync now, Exit completely.
- **Connections per download** – 1 (no splitting) to 16; more is usually faster.

Video player

- **Open video files in the Velivo player**, **Play right after opening**, **Resume where you left off** (this computer only), **Loop the video**. Details in chapter 8.

Torrents

- **Download torrents (magnet and .torrent) in Velivo** – off by default; **Torrent zone folder** (Choose… / Default), **Download** and **upload speed**, **Share after download** (ratio, 0 = not at all) and **for how long**, **Most peers**. Details in chapter 8.

Browser junk (cache)

- Page cache, compiled scripts and graphics cache – can be deleted without losing logins. **Junk folder** (e.g. on a RAM disk), **Default**, **Delete junk on every start**, **Currently uses: …**, **Clear junk now**.

Page loading speed

- **Faster page opening** – Velivo starts loading a page when you hover a link and pre-connects to servers of visible links (off in private and banking tabs; better off on a data cap).
- **Cache size** – automatic or a chosen size (after a restart).

Data

- **Clear browsing data now…** – tick: browsing history, cache, download history, cookies and sessions (logs you out), engine form and card data, passwords saved in the engine.
- **Per-site privacy → Privacy and anti-fingerprinting panel…** – domain rules: block JavaScript, send no cookies, force tracker blocking, clear data automatically on visit; a **trusted domains** list; a "What was blocked and why" log with full address and reason.

User profiles

- Active profile (e.g. Work, Private) and **Manage users/profiles…** – add, switch, delete, custom profile icon. Each profile has its own logins, history and settings.

Sync

- **E2E sync** – **Export / Import package…**: an encrypted file (password + AES-GCM) to move to another device.
- **Real-time sync (LAN)** – settings, bookmarks, passwords, privacy rules and Quick Access between home computers. **Pair a device on the network…** (compare a short code), **Save recovery file…** / **Restore pairing…** (password of at least 12 characters), **Quiet LAN mode** (no

pop-ups), **LAN diagnostic panel…** (devices, status, log).

22. Troubleshooting and where data is stored

Problem	What to do
A page blocks something it shouldn't	Privacy button → **Add to trusted**, or right-click → Restore blocked elements
A cookie banner is still shown	That banner has no "Reject" button. Velivo never clicks "Accept"
Computers don't see each other	Same profile on both, the firewall allows Velivo, automatic Windows time on both (a difference over 5 minutes blocks the exchange)
YouTube download fails	Wait a moment and try again. yt-dlp updates itself, and YouTube sometimes changes things
No icons in Quick Access	Open a new tab and wait a few minutes; " ⋮ " menu → Fill in missing icons
An offline voice doesn't read	Delete the `Piper` folder (below). It downloads again
Antivirus blocks the installer	The file has no paid code signature – add an exception
"An unexpected error occurred"	Velivo keeps running. Details are in `bledy.log` – attach it when reporting

Where data is stored

What	Where
Program	`%LOCALAPPDATA%\Programs\Velivo`
Settings, bookmarks, passwords, history, page memory	`%LOCALAPPDATA%\Przegladarka` (other profile: `Profiles\<name>`)
Offline voices	`%LOCALAPPDATA%\Przegladarka\Piper`
yt-dlp and FFmpeg	`%LOCALAPPDATA%\Przegladarka\narzedzia`
Error log	`bledy.log` in the profile data folder

All data stays on your computer. Velivo has no server and no account, and sends nothing about you.

Part II

Technical appendix – every feature step by step

Velivo 1.22 – Detailed guide (technical appendix)

This document describes every Velivo 1.22 feature point by point: what it is for, where to find it, how to use it step by step, how it works technically and what its limits are. It is based on the source code (`src/`). Where a feature works only partly or something is missing, it says so plainly.

Conventions: **Where** – the place in the interface. **Steps** – the order of actions. **Technically** – mechanism, files, parameters. **Notes** – limits and dependencies. Data file names refer to the profile data folder: `%LOCALAPPDATA%\Przegladarka` (default profile) or `%LOCALAPPDATA%\Przegladarka\Profiles\<name>`.

1. Architecture and data files

1.1 Engine and shell

- Velivo is a WPF program (.NET 10, Windows 10 1809+ x64) with the **Microsoft Edge WebView2** page engine. Pages are rendered by the same engine as Edge; the window, tabs, menus, privacy, downloads and sync are Velivo's own code.
- Engine profile: `Profil\` in the data folder. Banking mode has a separate engine profile.
- One program window: a second launch passes addresses to the running window through a named pipe (name depends on the data folder) and exits. Links from other programs open as new tabs.
- Velivo presents itself to sites as Microsoft Edge: the "Microsoft Edge WebView2" brand is removed from Client Hints headers and `navigator.userAgentData` (Google blocks sign-in in "embedded browsers"). Extension pages (`chrome-extension://`) see the real brand.

1.2 Files in the data folder

File / folder	Contents
`ustawienia.txt`	settings (key=value)
`lan-sync-key.dpapi`	LAN key encrypted with DPAPI (CurrentUser)
`zakladki.txt`, `zakladki-usuniete.txt`	bookmarks (address TAB title), list of deleted ones (90 days)
`historia.txt`, `historia-usuniete.txt`, `historia-wyczyszczona.txt`	history (time, address, title, session), deletions for sync
`sesja.txt`, `sesja.txt.aktywna`, `karty-przypiete.txt`	open tabs, active tab, pinned tabs
`zestawy-kart.json`	saved tab sets
`hasla.vault`	local passwords (DPAPI)
`autouzupelnianie.vault`	addresses, cards, accounts (DPAPI)
`bank*.json`	banking mode (fields encrypted with AES-GCM)
`prywatnosc.txt`, `prywatnosc-log.txt`	domain rules, block log
`ochrona-dozwolone.txt`	sites accepted as genuine after a fake-site warning
`ciasteczka-wyjatki.txt`	sites without cookie-banner rejection
`elementy.txt`	manually blocked elements (domain + CSS selector)
`powiekszenie.txt`, `tryb-stron.txt`, `widok-telefonu.txt`	zoom, dark/night mode, phone version – per site
`pamiec-stron.jsonl`	local "Where did I read that?" index
`pobrane.json`	download list
`skroty-wyszukiwania.txt`	search shortcuts
`Filtry\`	downloaded filter lists
`Dodatki\`	extensions, uBOL, sync list
`Miniatury stron\`	Quick Access thumbnails
`narzedzia\`	yt-dlp, FFmpeg
`sync-device-id.txt`, `sync-peers.json`, `lan-zmiana.txt`	sync
`bledy.log`, `debug.log`	error log, diagnostic log (`VELIVO_DEBUG=1`)

1.3 Launch parameters and variables

- `--profil <name>` – start with the chosen user profile.
- `--klawiatura` – open only the on-screen keyboard (the "Velivo Klawiatura" shortcut).
- `PRZEGLADARKA_DANE=<folder>` – a separate data folder (e.g. for tests); also a separate single window.
- `VELIVO_DEBUG=1` – logs window-closing events to `debug.log`.
- Safety net: an unhandled exception does not close the program; it goes to `bledy.log`, and a message appears at most once every 30 s for the same error.

2. Window, toolbar and navigation

2.1 Toolbar

Where: the top bar. Buttons (from the left): new private tab, new tab, Back, Forward, Reload, home page, address bar, profile, bookmark (star), privacy shield, dark mode, read aloud with speed, reader mode, screenshot, extensions, downloads, history, settings, "More tools".

- **Compact bar:** in a narrow window labels shrink and some buttons move to the "…" menu. The "Always compact toolbar" setting forces this layout.
- **Mouse wheel on buttons** changes values without clicking: on the night-mode button – strength, on the zoom button – percent, in the "Video on top" window – transparency.

2.2 Keyboard shortcuts

Shortcut	Action
Ctrl+T	new tab
Ctrl+Shift+N	new private tab
Ctrl+W	close tab
Ctrl+Shift+T	restore closed tab
Ctrl+Shift+A	search tabs
Ctrl+D	add/remove bookmark
Ctrl+H	history
Ctrl+J	downloads
Ctrl+O	open a file from disk (video or PDF)
Ctrl+Shift+F	"Where did I read that?"
Ctrl+Shift+U	read the page aloud / pause
Alt+← / Alt+→	back / forward
F5	reload
Ctrl + / Ctrl − / Ctrl 0, Ctrl+wheel	zoom

2.3 Mouse gestures

Where: Settings → Search and start → "Mouse gestures". **Steps:** hold the right button and move: ← back, → forward, ↑ new tab, ↓ close tab, ↓→ reload. A plain right-click opens the menu. **Technically:** a page script recognises the movement and sends a message to the program only with a random launch token; a page cannot fake a gesture.

2.4 Right-click menu on a page

Items (depending on context): search for the selected text, translate the selection into Polish, translate the page into Polish/English (Google Translate), read the selection aloud, "Read from here", reader mode and summary, "Where did I read that?", Video on top, Picture in picture, Download video, Block element, Restore blocked elements, Reject / Don't reject cookie banners on this site, Show the phone version / Back to the desktop version, Screenshot (visible part / whole page), Velivo tools (Privacy and anti-fingerprinting, Detect media, Download manager, LAN diagnostics, Extensions, History, Search tabs, profile switch).

3. Tabs and windows

3.1 Tabs – basics

Where: the tab bar; right-click a tab.

- Tab menu: Reload, Duplicate tab, Pin/Unpin, Mute/Unmute, Auto-refresh, Add to group, Tab sets, Send to…, Open as private, Close, Close others, Close tabs to the right, Restore closed (Ctrl+Shift+T).
- **Links in the same tab** (Settings → Tabs): links with `target=_blank` open in the current tab, so Back works. Ctrl+click and middle-click still open a new tab.

3.2 Pinned tabs

Steps: right-click a tab → "Pin tab". The tab moves to the start of the bar, loses its ✕ and comes back after a restart. **Technically:** `karty-przypiete.txt`; synced over LAN (missing pinned tabs are opened and pinned on the other computer).

3.3 Tab groups

Steps:

1. Right-click a tab → "Add to group" → "New group…" → type a name.
2. More tabs: "Add to group" → pick an existing one.
3. Clicking the group label collapses/expands it. Right-click the label: "Rename group…", "Color" (8 colors: blue, green, orange, purple, red, teal, pink, grey), "Ungroup (tabs stay)", "Close all tabs in the group".
4. "Remove from group" in the tab menu.

Technically: groups are saved with the session (tab position, name, color, collapsed state) and come back after a restart.

3.4 Tab sets

Steps: tab menu → "Tab sets" → "Save open tabs as a set…" → name (e.g. Work). Later: "Tab sets" → name → "Open all tabs" or "Replace current tabs"; "Delete set". **Technically:** `zestawy-kart.json`.

3.5 Auto-refresh

Steps: right-click a tab → "Auto-refresh" → Off / 1 / 5 / 15 / 30 min. The tab shows a badge with the tooltip "Refreshing every N min". **Notes:** the timer works per tab and is not saved across restarts.

3.6 Searching tabs

Steps: Ctrl+Shift+A → type part of a title or address → choose the tab.

3.7 Restoring the session and closed tabs

- Settings → Tabs → "Restore tabs from the previous session on start". Private tabs are never saved. With "Clear data on close" the session is not saved either.
- Ctrl+Shift+T restores recently closed tabs (an in-memory stack).

3.8 Private tabs

Steps: Ctrl+Shift+N or the private-tab button; from the tab menu "Open as private (incognito)". **Technically:** a separate, isolated WebView2 store (InPrivate): no history, session or content memory; skipped by "Where did I read that?".

3.9 Muting a tab

Right-click a tab → "Mute tab" / "Unmute tab". A tab playing sound shows a speaker badge.

3.10 Zoom

- Default for all sites: Settings → Look. For a single site: Ctrl+wheel, Ctrl ±, Ctrl 0 or the wheel on the percent button.
- **Technically:** `powiekszenie.txt` (host → percent); a change carries over to other tabs with the same site.

3.11 Phone version of a site

Steps: right-click the page → "Show the phone version of this site". Back: "Back to the desktop version of this site". **Technically:** the tab presents itself as an Android phone (User-Agent and Client Hints, which Google among others checks) on every navigation; the choice is remembered per site in `widok-telefonu.txt`.

3.12 Screenshot

Steps: the camera button or right-click → "Screenshot" → "Visible part of the page" or "Whole page (with scrolling)". **Technically:** PNG saved to `Pictures\Zrzuty Velivo` and copied to the clipboard; a bubble with "Open" and "Show in folder" buttons. On bank and payment sites a screenshot is impossible (see 7.8).

3.13 Page entrance effects

Where: Settings → Look and readability → "Content entrance effect" and "Entrance effect speed".

- Effects: **Focus-in** (default – content emerges from a blur), **From darkness (cinematic)**, **Gentle dimming**, **None**.
- Speed: slider 0–5000 ms in 100 ms steps; 0 = automatic.

Technically: the effect applies to the whole page window (including the background and pinned menus); after clicking a link the page dims slightly and the new content appears with the effect.

3.14 Faster page opening

Where: Settings → Page loading speed. **Technically:** hovering a link on the same site starts loading it in the background; Velivo connects in advance to the servers of visible links. Off in private and banking tabs. Better switched off on a data cap.

4. Bookmarks and history

4.1 Bookmarks

- Ctrl+D or the star – add/remove a bookmark for the current page. The "All bookmarks" button – the list; right-click a bookmark: rename, delete.
- **Technically:** `zakladki.txt`; deletions go to `zakladki-usuniete.txt` (90 days) so LAN sync removes them on the other computer too. Adding again clears the entry from the deleted list.

4.2 History (Ctrl+H)

Layout: day → session (one browser launch, "● Current session") → pages. **Steps:** the "Search titles and addresses" field; double-click opens a page; right-click: "Remove from history", "Remove found entries from this day", "Remove the whole day from history", "Open all in new tabs", "Remove the session from history"; the "Clear all history" button. **Technically:** `historia.txt` (time TAB address TAB title TAB session); old entries without a session number are split at 30-minute gaps. Saving can be turned off: Settings → Privacy → "Save browsing history".

5. Look and personalisation

5.1 Style and themes

- Settings → Look → "Look style": **Modern** (single-color Windows 11 Segoe Fluent icons, one blue accent, color only on hover or when on) or **Colorful** (colored buttons and emoji).
- "Browser theme": Light, Graphite, Navy, Night violet, Forest, Ocean, Sunset, Black (OLED), Paper, Mist.

5.2 Dark and night mode for pages

Where: the moon button; Settings → Look. **Steps:** a click switches light → dark → night. In night mode the mouse wheel on the button sets the strength 5–100%. **Technically:** dark mode uses the engine's dark theme: sites with their own dark look switch to it, others are darkened, and photos keep their colors. Night mode is a warm, semi-transparent layer over the page that does not catch clicks. The mode is remembered per site (`tryb-stron.txt`). If the engine started in another mode, Velivo fixes the page with CSS or offers a restart (tabs come back).

5.3 Language

Settings → Look → "Interface language": Automatic (from the installer / Windows), Polish, English. The change takes effect after a restart. All windows are translated, including those built in XAML, and Quick Access.

5.4 Search engine and search shortcuts

- Settings → Search and start → "Search engine in the address bar" (DuckDuckGo, Startpage – default, Brave Search…).
- "Search shortcuts…": word + space + query, e.g. `yt cats` searches YouTube. Add your own in the dialog; saved in `skroty-wyszukiwania.txt`.

5.5 Home page and new tab

Settings → Search and start → "Home page" and "Quick Access as the new tab page".

5.6 Velivo user profiles

Where: the profile button on the toolbar; Settings → Profiles → "Manage users/profiles…". **Steps:** "Add user/profile…" → name (letters, digits, `-`, _); "Profile icon" → "Save icon"; "Switch to selected" (Velivo restarts in that profile); "Delete selected profile" (the active or default profile cannot be deleted). **Technically:** each profile has its own data folder `Profiles\<name>`: sign-ins, history, passwords, settings and Quick Access. The active profile is stored in `active-profile.txt`.

5.7 Sound – Velivo speakers

Where: Settings → Security and downloads → "Velivo speakers".

- Pick a specific output instead of "Default Windows output".
- **"Don't lose sound":** every 2 s Velivo checks whether the chosen output is free. When a music program (Ableton, Cubase – ASIO/WASAPI exclusive) takes the card, Velivo's sound moves to another active output and returns when it is released.
- " Windows volume mixer…" opens Windows settings, where you can pin speakers permanently for "Velivo" / "Microsoft Edge WebView2".

Technically: the MMDevice API and the undocumented `IAudioPolicyConfigFactory` interface (the same one EarTrumpet uses) to assign the app's output.

5.8 Default browser

Settings → "Set Velivo as the default browser…". Velivo registers with Windows (class `VelivoHTML`, `StartMenuInternet\Velivo`) and opens "Default apps". Windows does not let a program set itself – you have to click "Set default".

5.9 Working from the tray

Settings → "Stay in the tray when the window is closed". Closing the window hides Velivo to the icon by the clock. Background pages are paused (a video in the "Video on top" window keeps playing). LAN sync keeps working. Right-click the icon: Open Velivo, Sync now, Close Velivo completely. The icon pulses while syncing.

6. Quick Access (new tab page)

Quick Access is a built-in extension (`chrome-extension://…/kod/newtab.html`).

6.1 Shortcuts and groups

Steps: "+ Shortcut" → address, name, optionally an image ("Choose image…" from a file, paste an image, "Remove custom") → Save. "+ Group" – a new group (e.g. Start, Finance, Music). A shortcut can be edited and deleted with the icons on its tile. From any page: right-click → "Add to Quick Access".

6.2 Page thumbnails

Menu ⋮ → "Page thumbnails". An extension in WebView2 cannot capture a tab, so Velivo does it: an invisible off-screen window opens the pages one by one without history, sound, pop-ups or downloads and saves thumbnails in `Miniatury stron\`.

6.3 Quick Access profiles and PIN

Profile list → "+ New profile…". Menu ⋮ → "PIN for this profile" – the PIN is protected with a salt and a hash. Entering a profile with a PIN requires the code, and so does adding a page to it. "Delete this profile".

6.4 Bin

Deleted shortcuts go to the bin: menu ⋮ → "Bin…" → "Restore" or empty it.

6.5 Look

Background themes and tile size.

6.6 Backup, folder and import

- Backup to disk ("Quick Access backup") and a data folder handled through the bridge (SejfMost) – "Extension folder".
- Import bookmarks from a JSON or HTML file. Backup limit: about 100 KB, i.e. about a thousand shortcuts.
- "Name of this computer" – to tell backups apart.
- In Chrome the same extension can sync with a Google account ("Sync with Google account", "Sync OK" state); in Velivo it is synced over LAN (see 10.8).

7. Privacy and security

7.1 Velivo ad blocking

Where: the shield on the toolbar; the "Turn AdBlock on/off" button; Settings → Ad blocking. **Technically:** a domain filter in a `HashSet` (checked by successive host labels) + a list of address fragments. From EasyList-format lists the rules that block whole domains (`||domain^`) are used. Your own rules: `filters.txt` in the data folder. A blocked request gets a 403 response.

7.2 Full filter lists

EasyList, EasyPrivacy and the Polish list (about 97k rules). Downloaded in the background and refreshed every 4 days; "Update lists now" downloads immediately. The file is replaced only after a successful download. The filter is built in the background (a few hundred ms) and swapped as a whole. On error: "Could not download N list(s) – using the previous ones".

7.3 uBlock Origin Lite

Built in (GPL-3.0), on by default, can be turned off; "uBlock Origin Lite settings...". The installer includes the latest release. Once a week Velivo checks for a new release on GitHub (`uBlockOrigin/uBOL-home`) and installs it **only if the SHA-256 matches the digest given by the GitHub API**. uBOL blocks (`net::ERR_BLOCKED_BY_CLIENT`) are counted on the shield without double counting Velivo's blocks.

7.4 Shield and block log

- The shield shows the number of blocked items on the page and the "Blocked on this page" list.
- The "Privacy and anti-fingerprinting" panel → "What was blocked and why": time, domain, reason (tracker – domain rule / AdBlock, cookies removed by a rule...), the full address after selecting an entry; "Clear the block panel". Saved in `prywatnosc-log.txt`.

7.5 Tracking prevention

Settings → Privacy → "Tracking prevention":

- **Balanced** (recommended): known trackers are blocked, embedded content (posts from X, videos) works.
- **Strict:** also blocks embedded social-media content. On trusted domains it works like balanced unless you tick "Force tracker blocking" for the domain.

Technically: the engine's protection level is shared by all tabs, so Velivo sets it according to the active page.

7.6 Privacy rules per domain

Where: shield → "Privacy and anti-fingerprinting panel"; Settings → per-site privacy. **Steps:** type a domain (you can paste a full address – the domain is kept) → tick: "Block JavaScript for the domain", "Don't send cookies for the domain", "Force tracker blocking for the domain", "Automatically clear data after visiting the domain" → "Save rule". "Delete rule". The "Trusted domains" list: "✔ Selected domain to trusted" (from the block log). **Technically:** `prywatnosc.txt`; rules apply only to web pages (not extensions). Synced over LAN.

7.7 Privacy receipt and fingerprinting

Where: click the shield (Settings → "Privacy receipt on the shield").

- Shows: how many outside companies and how many countries the page connected to, data brokers, the most frequent companies and **attempts to identify the computer**: canvas (reading an invisible canvas < 250,000 px), WebGL (reading the graphics card vendor/model – parameters 37445/37446), audio (OfflineAudioContext).

- **Notes:** Velivo **detects and reports** fingerprinting but **does not change the values read** (no "farbling" like Brave and no normalisation like Firefox). Protection comes from tracker blocking via the lists and hiding the WebView2 brand.

7.8 Safe payments (screen-recording block)

Settings → Security → "Safe payments". On bank and payment sites (PayPal, Monzo, Revolut, Barclays, HSBC, Lloyds, NatWest, Santander, PKO, mBank, Stripe and others) the Velivo window gets `SetWindowDisplayAffinity(WDA_EXCLUDEFROMCAPTURE)` – recording programs and screenshots see an empty area. A message appears. You cannot take a screenshot on these sites.

7.9 Fake-site detection (offline)

Settings → "Detect fake bank, shop and portal sites". Before opening a page Velivo checks:

1. **A brand on a foreign domain** – a list of protected brands with their real domains (PayPal, eBay, Amazon, Facebook, Instagram, WhatsApp, Google/Gmail, Microsoft/Outlook, Apple/iCloud, Twitter/X, Netflix, Spotify, banks and others); e.g. `paypal-secure-login.com`, `ebay.co.uk.konto.top`.
2. **Typos and swapped characters:** `paypa1.com`, `rnicrosoft.com`, `amaz0n.com`.
3. `xn--` **(punycode) addresses** with letters from other alphabets (homoglyphs).
4. **Fakes of sites you have saved passwords for.**

A warning "Careful – this may be a fake site!" appears with **No** as the default answer. "Yes" remembers the site as genuine (`ochrona-dozwolone.txt`).

7.10 HTTPS, SmartScreen, pop-ups

- "Always use an encrypted connection (HTTPS)" – unencrypted sites open only after a warning.
- "Warn about dangerous sites and files (SmartScreen)" – addresses are checked with Microsoft.
- "Block pop-ups opened without a click".

7.11 "Do Not Track" and cookie banners

- "Send Do Not Track signals" – DNT and Global Privacy Control headers.
- "Automatically reject cookie banners (GDPR)" – Velivo clicks "Reject" / "Only necessary"; if there is no such button, it clicks nothing (never "Accept"). Exception: right-click → "Don't reject cookie banners on this site" (`ciasteczka-wyjatki.txt`).

7.12 Blocking page elements

Steps: right-click → " Block element (ad)..." → hover – the element is highlighted; the wheel or arrows change the scope (parent/child element); a click or Enter blocks; Esc cancels. Undo: right-click → "Restore blocked elements on this site". **Technically:** a rule = domain + CSS selector in `elementy.txt`; the element gets `display:none` on every visit.

7.13 Shop pressure-trick detector

Settings → "Warn about pressure tricks in shops". Detects fake countdown timers, "last items", "X people are watching", pre-ticked extras (Velivo **unticks** them) and hidden fees; a message appears above the page.

7.14 Page ↔ program channel

Pages have no access to `chrome.webview` (hidden when the document is created). Velivo scripts keep the channel before hiding it and sign messages with a random token of this launch; a message without the token is ignored. Quick Access accepts commands only from its own address.

7.15 Downloaded files – "file from the internet" mark

Every file downloaded by the Velivo manager, yt-dlp or the engine gets a `Zone.Identifier` stream with `ZoneId=3` (Internet) and `HostUrl=about:internet` (no page address – privacy). SmartScreen warns on launch, and Office opens the document in Protected View. An existing mark is not overwritten; on FAT32/exFAT drives the mark cannot be written.

7.16 Clearing data

- "Clear data on close (history and cache)" – does not sign out of accounts.
- "Clear browsing data now…": Browsing history, Cache, Download history, Cookies and active sessions (signs out of accounts), Form and card data saved by the engine, Passwords saved in the engine.

7.17 Program resilience

- An error in the program does not close the browser (see 1.3).
- Protection against the whole window being closed by the WebView2 control's internal handling (`window.close` on a page closes only the tab).

8. Passwords, Sejf and autofill

8.1 Velivo password manager

Where: Settings → Privacy → "Local password manager…". **Steps:** "Search by domain, login or name"; "Add" (domain, user, password, name, notes; "Generate a strong password", "Paste from clipboard", "Show/hide password"); "Edit"; "Copy login", "Copy password"; "Delete selected"; "Delete all saved passwords". Columns: Name, Domain, User, Source, Changed. **Technically:** `hasla.vault` encrypted with DPAPI (CurrentUser) with its own entropy. Generator: `RandomNumberGenerator`, at least one lowercase, uppercase, digit and symbol each, then a Fisher–Yates shuffle.

8.2 Saving and filling passwords

- "Offer to save passwords": after signing in, the question "A password was detected on … Add it to the Velivo database?" or "A new password was detected for … Update the entry?".
- Filling: "Fill in the saved login and password from Velivo" / "Choose an account to fill in" / "Another account from the Velivo database…".
- **Security:** an entry is matched **only to the real domain** – no matching by name, so a fake site won't get the password.

8.3 Import and export

- "Import passwords CSV…": Chrome, Edge, Firefox, KeePass, KeePassXC (entries from the bin are skipped); quoted fields with new lines are supported.
- "Export passwords CSV…" – **a plain-text file**, store it carefully.
- " From Chrome / Edge / Brave / Opera" (Settings → Import): bookmarks are read directly from those browsers' files (Vivaldi too); passwords – Velivo opens the browser's password page for a CSV export.

8.4 Sejf (external manager)

Settings → "Logins from Sejf: key on the toolbar on sign-in pages". When a page has a password field, Velivo asks Sejf (the `SejfMost.exe` program, a registered native host) for logins for **the domain from the tab's real address**. Clicking the key fills the form; before filling Velivo checks that the tab is still on that site. Sejf may ask to be unlocked (Sejf window, 1 minute limit). From the extension popup you can also save a login to the encrypted vault. Without Sejf: "(Sejf not found)".

8.5 Form autofill

Settings → "Form autofill (addresses and cards, local encrypted database)".

- Data: **address** (first name, last name, street, postcode, city, phone, e-mail), **card** (no CVC), **bank account** (IBAN, account number, sort code).
- Clicking an empty field fills the whole form on any https site. Velivo fills in the card and account only after you confirm the site address.
- Saving always happens only after asking ("A filled-in address form / payment card details were detected").
- **Security:** only fields really visible to a person are filled; hidden fields never are (a site can't extract data with a hidden field); only real clicks count.
- "Show saved data…", "Delete saved cards", "Delete saved addresses". File `autouzupelnianie.vault` (DPAPI).

9. Banking mode

9.1 What it is

A separate, isolated browser profile for banks, payments and shopping, with its own encrypted database. Its own cookies and sign-ins, no extensions and no history. Normal tabs see nothing of it and vice versa. Banking tabs are green and show a icon.

9.2 First start

1. Click the green button on the tab bar.
2. Set a password (min. 8 characters) and repeat it.
3. Optionally: tick "Also require a hardware key (YubiKey, Google Titan…)", insert the key, click " Add key (insert it and touch it)". Windows asks for a touch **twice** (registration + reading the secret). Add all your keys (a spare!) in the same dialog. Windows finds NFC/Bluetooth keys itself after clicking "Add".
4. Choose the idle lock time and click "Save".

9.3 Opening and locking

- Key in the port → Velivo asks for a touch right away; no password needed. No key → password.
- Automatic lock after inactivity (default 10 min, choice 1–60 min) closes all the mode's windows. " Lock now" – immediately.
- After the mode is closed, the banking profile's cache and history are cleared (sign-ins and "remember me" stay).

9.4 Encryption (technical)

- Password → PBKDF2-HMAC-SHA256, **600,000 iterations**, 16 B salt → 32 B key.
- Data (cards, notes, sites, documents, accounts, bills, logins, payment history) encrypted with **AES-256-GCM** (random 12 B nonce, 16 B tag) by the `SealList` function.
- Hardware key with the **FIDO2 hmac-secret** extension: the secret from the key goes through HKDF-SHA256 ("velivo-bank-kk") and becomes the second part of the key (`SHA256(Kp + Kk)`). Without the physical key the password alone is then not enough. Key signature: ES256; WebAuthn via Windows `webauthn.dll`.
- File `bank.json` (profiles: `bank-<name>.json`), read into RAM only when the file changes.

9.5 Banking-mode menu (right-click)

Search my database · My logins and passwords · Confidential data (documents) · Bank accounts · Bills to pay · Bills sheet (in a tab) · My banks · My online shops · Add this site to My banks / shops · My cards · Fill in the card on this page · My notes · Fill in the login from a note · Fill in selected characters · Copy the login form description · ⚙ Banking mode settings · Opening log · Database backup · Restore database from backup · Banking profiles / New banking profile · Lock now · Banking mode guide · I forgot the password – clear banking mode.

9.6 My banks and My online shops

- A list of your sites; a click opens the site in banking mode.
- Adding: on an open banking tab " Add this site to My banks/shops" or on the list " Add a bank / shop (name and address)…".
- **Protection:** when you open a bank or shop from the list in a normal tab, "Switch to banking mode" / "Stay here" appears above the taskbar (recognised by the main domain, including other addresses of the same company). When, with the mode open, you visit a site resembling your bank, a big warning appears.

9.7 Bank login details and selected characters

- "✏ Login details" (in My banks): login / customer number, passcode / PIN, password, memorable information – separately for each bank. Enter the **full** password and passcode.
- " Fill in login" – fills the login and password on the bank's site.
- " Fill in selected characters" – when a bank asks e.g. for the 2nd, 5th and 9th character (RBS, NatWest, Bank of Scotland, TSB, Lloyds, Halifax), Velivo reads the numbers from the form itself and enters the right characters, including in drop-down lists.
- " Copy the login form description (without your data)" – if filling doesn't work, it copies only the field description for a report.

9.8 My cards

" My cards" → click a card to view and change its details. Fields: name (e.g. "Visa PKO"), number (validated), valid until (MM/YY), cardholder name, CVV (optional, 3–4 digits, encrypted). " Show number and CVV". " Copy number / date / CVV" – the clipboard clears after 30 s. " Fill in the card on this page". On opening the mode – a reminder about cards expiring soon.

9.9 My logins and passwords (in the mode)

Name, login, password (hidden, " Show hidden data"), site address, note. " Open site". " Fill in on the open page" – **only when the open tab has the same domain** (a message otherwise). " Generate password" – 20 characters, copied, clipboard cleared after 30 s.

9.10 Confidential data (documents)

Type (ID card, Passport, Driving licence, PESEL, NI number, Health insurance, Other document), full name, number (hidden), valid until (DD.MM.YYYY), note. A reminder about documents expiring within 60 days.

9.11 Bank accounts

Name (e.g. "My account", "Rent – landlord"), owner, number / IBAN (hidden), sort code / BIC (SWIFT), bank, transfer reference / note. buttons by each field – error-free copying into a transfer.

9.12 My notes

Categories: Login, PIN, Transfers, Recovery codes, Other. Title + content → "Save". " Generate password", " Numbered characters" (the password character by character with numbers – helps with "enter the 3rd and 7th character"), copying. Note and search windows don't block the page (you can copy into a form). " Fill in the login from a note" – the note must contain `login: …` and `hasło: …` lines.

9.13 Bills to pay

Fields: What for (a list of names from the sheet), Amount, Due date (DD.MM.YYYY), Repeat (Weekly, Every 2 weeks, Every 4 weeks, Monthly, Every 2 months, Quarterly, Every half year, Yearly, One-off or your own, e.g. "co 10 dni", "every 3 weeks"), Remind days before, Customer number / reference, Payment page, Note. **Steps:**

1. Fill in the fields → " Add as new" (or "✔ Change selected", "Delete selected", "Clear fields").
2. After paying: select the bill → "✔ Paid (next due date)". The payment goes to the history (month = the due month), the due date moves on according to the repeat; a one-off bill disappears.
3. " Open payment page".
4. " Add bills from the sheet" – for each sheet column without a bill: the amount from the last month, monthly, due on the 1st of the next month. Correct the due day.
5. Finally "Save" – only then do changes and payments go into the encrypted database.

Reminders: on opening the mode – bills due within a few days and overdue ones, every day until paid. **Notes:** Velivo does not connect to the bank; you mark payments yourself. Two "Paid" in the

same month add up.

9.14 Bills sheet

Where: " Bills sheet (in a tab)" in the menu or in "Bills to pay".

- Row = month, column = bill; the "Total" column = the month's sum, the "Total" row = the bill's sum, bottom right = the grand total. Calculated live.
- Enter the month in words or numbers: "październik 2026", "October 2026", "10.2026", "2026-10".
- " Row (month)" – the month after the latest in the table, copying amounts from the last row.
- " Column (bill)" – name in the header; × under the name deletes the column, × at the end of a row deletes the month.
- " Load from Excel (CSV)" – Excel: File → Save as → CSV. Layout: 1st column = month, 1st row = bills. The ;, , or TAB separator is detected automatically. "Razem/Suma/Total/Sum" columns and rows are skipped. Existing months are filled in (empty cells don't erase amounts), new bills are added as columns, rows are sorted chronologically.
- "Currency you enter amounts in": £, zł, €, $, CHF, kr, Kč, Ft, lei, ₴, ¥ – **only a symbol, no conversion**; by default from the Windows region.
- " Save and close" – encrypted in the database and the tab closes; rows without amounts and without a valid month are skipped (with a notice). "Cancel" – no saving.
- **Technically:** the sheet page is loaded locally (no internet) with a one-time random token; Velivo accepts data only from that page and only if the tab's address hasn't changed.

9.15 Payment summary

The " Payment summary" window: a table of months and bills, adding a payment manually (bill, month, amount), "Delete this bill's payments for the month", " Load from Excel (CSV)...", " Save to Excel (CSV)" (export `Velivo-rachunki.csv`, UTF-8 with BOM, ; separator), currency choice.

9.16 Banking profiles, log, backups

- " Banking profiles" / " New banking profile (another user)..." – a separate mode for another person: their own password or key, separate sign-ins and data.
- " Opening log" – when, on which computer and how the mode was opened (wrong passwords too).
- " Database backup..." / " Restore database from backup..." – a `.vbank` file with all profiles, still encrypted (e.g. on a USB stick).
- "I forgot the password – clear banking mode..." – without the password and key the data cannot be read; this option deletes the mode and its data on this computer.
- Sync: all banking profiles go to paired computers in an encrypted package; bank sign-ins (cookies) stay separately on each computer.

10. Home network sync (LAN)

10.1 Principle

No cloud and no account. Computers with the same Velivo profile on one network exchange data directly (UDP, port 41919; Quick Access images – TCP on the same port). The Windows firewall must allow Velivo. On by default (Settings → Sync).

10.2 Before pairing

Every 10 s Velivo sends **only** an `announce` packet (identifier, computer name, profile) and **neither sends nor accepts any data**. The log says: "LAN sync waiting for pairing – no data is sent". When another Velivo with the same profile is on the network, one of the computers (the smaller identifier) proposes a connection.

10.3 Pairing step by step

1. Turn sync on and save the settings on both computers.
2. On one: "Pair a device on the network..." (or accept the proposal).
3. The same **6-digit code** appears on both screens. Compare it and confirm on both.
4. On the first connection choose whose **settings** stay (bookmarks, passwords and Quick Access are merged anyway).

Technically: ECDH P-256 key exchange; transport key = HMAC-SHA256(agreed secret, "Velivo LAN pairing v1" + pairing id); code = HMAC(key, ...) mod 1,000,000; HMAC confirmation; the sync key (32 random bytes) is sent encrypted with AES-GCM. Requests from another profile are rejected, pairing attempts expire.

10.4 Encryption after pairing

Key → PBKDF2-SHA256 (250,000 iterations) → 32 B encryption + 32 B authentication. Packets: AES-256-GCM with associated data (type, sender, device, profile, time, hash), "hello" packets signed with HMAC-SHA256 (constant-time comparison). GZip compression before encryption, an 8 MB limit after decompression, 60,000 B per UDP packet. Packets older or newer than ±5 min from the receiver's clock are rejected (replay protection) – set automatic Windows time on both computers.

10.5 What is synced

Data	How
Settings	the newer change wins; local ones are not transferred: speakers, folders (downloads, videos), language, reader window and video window size, LAN key, quiet mode
Bookmarks	merged from both computers; deletions via the deleted list (90 days)
Passwords	merged (only in an encrypted package)
Privacy rules, profiles	the newer change wins
Extension list	missing store extensions are installed automatically
Pinned tabs	opened and pinned
History	merged, separate packages; deletions and "clear all" are transferred
Quick Access with images	archive ≤24 MB over TCP, only from a paired device, with path protection inside the archive
Banking mode (all profiles)	encrypted package; bank cookies stay local
Open tabs	**no** – each computer has its own (use "Send to..." to move one)

10.6 Profile isolation

Only computers with the same profile sync. Packets from another profile are skipped; when the other computer switches to a different profile, Velivo asks whether to switch too, to keep syncing.

10.7 Send a tab to another computer

Right-click a tab → " Send to…" → choose the computer. The tab opens on the other computer; for a YouTube video also from the same moment. The package is encrypted.

10.8 Indicators, notifications and diagnostics

- The tray icon pulses while sending and receiving changes.
- Bubbles " Synced with Velivo on the local network" / "Synced with … (settings accepted)". "LAN quiet mode" turns the bubbles off (the log keeps working).
- "LAN diagnostic panel…" (also Velivo tools → LAN sync diagnostics): status ("off", "active but no connected devices", "connected to N devices"), detected devices, packet and error counters, last receive/send time, log (250 entries), "Refresh now", "Clear log".
- Warnings: clock difference ("Rejected packets from …: clock differs by N min"), packet too big, start/receive/send errors, incomplete state.

10.9 Recovery file and file-based sync

- "Save recovery file..." – password min. 12 characters (twice); file ≤1 MB, PBKDF2 600,000 + AES-GCM. "Restore pairing…" after a reinstall restores the key without pairing again.
- **E2E sync by file:** "Export package…" / "Import package…" – settings, bookmarks, passwords and other data in a password-encrypted file (PBKDF2 600,000 + AES-GCM) to move e.g. on a USB stick.

11. Video

11.1 Video on top

Where: the ▣ button when hovering over a video; right-click → "▣ Video on top (small window)".
Steps and window elements:

1. The window shows just the video (the rest of the page is hidden); on first use it appears by the right edge of the screen.
2. Drag it by the top bar; resize with the frame (min. 160×114).
3. Pin: "always on top" on/off (blue = pinned).
4. Transparency 15–100%: slider and mouse wheel – e.g. 40–60% over a document.
5. A video seek bar at the bottom of the window.
6. "Back to tab" – the video returns to the tab.

Technically: a separate WebView2 view; transparency via a Windows window layer. Position, size, pin and transparency are remembered (with a check that the window fits on screen). YouTube: resumed through its player and the time in the address (works after a Velivo restart too). Closing the tab doesn't stop the video; closing the window does. **Watching with the browser closed:** the video window is a separate window. You can close the tab and even the whole main Velivo window – the video keeps playing and the program runs only for the window (LAN and Sejf connections are then stopped). When you start Velivo again while only the video window is playing, Velivo closes it and starts the full window. The "Back to tab" button starts the browser on the same page and at the same moment in the video.

11.2 Picture in picture (PiP)

The button when hovering over a video (Settings → "Picture in picture button over videos") or right-click → " Picture in picture" (the largest video). The PiP window stays even after the tab is closed. Videos in frames (e.g. embedded YouTube) get the button on hover.

12. Downloads

12.1 Download manager (Ctrl+J)

Velivo downloads files itself (like Internet Download Manager):

- **Multiple connections:** Settings → "Connections per downloaded file" 1–16. Segments are used for files ≥2 MB when the server supports ranges (Range).
- **Pause / Resume** – from where it stopped; if the server doesn't allow it, a notice "will start over".
- **Retrying:** up to 5 attempts with growing delay after a dropped connection; no data for 60 s = error; "Too many redirects".
- **Signed-in files:** Velivo passes the page's cookies and follows redirects itself (cookies counted per host).
- **Fallback:** a file created by a page script cannot be taken over – the engine downloads it (`pobieranie-silnik.log`).
- "Ask where to save each downloaded file"; "Open", "Show in folder", "Cancel" (deletes the partial file), "Open the Downloads folder". The list is in `pobrane.json`.
- Edge's built-in download bubble is hidden.
- Closing Velivo with active downloads – a question.
- Every downloaded file gets the "file from the internet" mark (7.15).
- **Torrents** (Settings → Torrents, off by default): magnet and `.torrent` via aria2 (downloaded once, with consent, SHA-256 checked), `.torrent`/`magnet:` association in the installer, a separate zone `Downloads\Velivo-Torrenty` or a chosen folder, speed and seeding limits; buttons "Open folder", "Stop seeding", "Show in folder", "Delete file", "Remove from list"; after an interruption a question about deleting the partial files. From the tray they keep downloading after the window is closed; disabled in banking mode.

12.2 Video and audio downloads

Where: the ⬇ Download button when hovering over a video (Settings → "Download button over videos"), right-click → "⬇ Download video...", Tools → "Detect media to download". **Modes: Best quality** (video ≤1080p MP4 + M4A audio, merged with FFmpeg), **Simple** (one file without FFmpeg), **Audio only M4A**, **Audio only MP3** (FFmpeg, highest quality). The video folder is remembered. **Technically:** a plain video file → the Velivo manager; YouTube, streams (m3u8/mpd/blob) and other services → **yt-dlp** (downloaded once, about 18 MB, from GitHub; `-U` update every 14 days). **FFmpeg** (about 140 MB) is downloaded once when first needed. Both tools are checked with **SHA-256** from the same release's checksum file; a mismatching file is deleted and never run. The page address is passed after the `--` separator. **Notes:** subtitles are not downloaded with YouTube videos – you download subtitles for films in the player (18.3). In "Detect media" mode a `blob:` source is not a file and cannot be downloaded directly.

12.3 Detect media to download

A list of audio/video sources on the page with "Download", "Download as audio", "Show" buttons. The message "No audio/video sources detected" when there are none.

13. Reading

13.1 Reader mode ("Reader" button)

The article without ads and menus. Look: ☀ Light, Dark, Night with a strength slider. The reader window size is remembered. **Local summary:** no AI and no cloud – Velivo picks the most important sentences (weighted by frequent keywords of the text, title words and a bonus for the start of the article).

13.2 Read aloud (Ctrl+Shift+U)

- The whole page or a selection; right-click → " Read from here".
- The paragraph being read is highlighted and scrolled to the middle.
- The speed button (a click changes the pace), pause/resume, "Stop reading".
- Voice: Settings → Reading: "Automatic (page language: Polish/English)", Windows voices, natural online voices.

13.3 Natural offline voices (Piper)

Settings → Reading → the "Piper" voice. On first use Velivo downloads the Piper program (version 2023.11.14-2, SHA-256 checked) and the chosen voice (about 60 MB) to `%LOCALAPPDATA%\Przegladarka\Piper`; after that it works offline. The Piper process runs in the background; the page splits the text into sentences and highlights them.

13.4 "Where did I read that?" (Ctrl+Shift+F)

Settings → "Remember the content of pages you read". Velivo stores page text **only on this computer** (`pamiec-stron.jsonl`) and finds an article by words remembered from its content, not just the title. Skipped: private tabs, banks, payments, e-mail, pages with a password field.

13.5 Translation

Right-click: "Translate the selection into Polish", "Translate the page into Polish / English" (foreign → Polish, Polish → English) via Google Translate.

14. Extensions

14.1 Chromium extensions

Where: the Extensions button (developer mode).

- An extension icon bar with popups, "Open the extension popup", "Open the popup in a tab", "Extension settings page", pinning to the toolbar (`dodatki-odpiete.txt`), "Reload", removal.
- Loading an unpacked extension manually: "Extension folder (with manifest.json)".
- On start Velivo reloads extensions from a folder whose code changed (WebView2 keeps the old background script in its cache).

14.2 Chrome Web Store

On an extension's store page there is a " Add to Velivo" button (or "Install from link/ID..."). WebView2 has no store API, so Velivo intercepts the `.crx` download (CRX2 and CRX3), unpacks and installs it. The extension list syncs over LAN.

15. Performance and cache

15.1 Browser junk and a RAM disk

Settings → "Browser junk (cache)":

- "Junk folder" – e.g. a RAM disk (`R:\`). Velivo creates a `Velivo-smieci` subfolder there and clears **only its contents**. No links or external tools are needed. The change takes effect after a restart.
- "Default" – back to the profile.
- "Delete junk on every start" – pages load a little slower the first time.
- "Currently taking: …", "Clear junk now".

Technically: the page cache and compiled JavaScript go to the chosen folder (engine arguments). The graphics cache (`GPUCache`) always stays in the profile – it is cleared on the next start. Junk is cleaned only when no other Velivo window is running. **RAM disk benefits:** faster loading from cache, fewer writes to the SSD, the cache disappears when the computer is turned off.

15.2 Other mechanisms

- "Cache size (after restart)": Automatic or a chosen size.
- Request interception skips images and fonts (each intercepted request goes through the window thread); image ads are blocked by uBOL inside the engine at no cost to the window.
- The ad filter is built in the background and swapped as a whole.
- The banking profile is read from disk only when the file changes.
- Faster page opening (3.14).

16. Installing, updating, uninstalling

- The `Velivo-Setup-1.22.exe` installer (Inno Setup): language choice (Polski / English), installs to `%LOCALAPPDATA%\Programs\Velivo` without administrator rights, a Start menu shortcut and optionally a desktop one, registration as a browser, the "Velivo Klawiatura" shortcut.
- If it finds data from an earlier installation: "Keep my settings and data (update)" or "Clean install" (old data moved to a backup with a "kopia-..." suffix).
- .NET 10 is included in the installer (about 60 MB). If WebView2 Runtime is missing, the installer downloads it from Microsoft and installs it silently.
- The "Video player" option downloads the VLC libraries (about 145 MB, official VideoLAN package, SHA-256 checked) – only if you tick it.
- Silent install: `Velivo-Setup-1.22.exe /VERYSILENT /SUPPRESSMSGBOXES /NORESTART /LANG=en /DIR=<folder>`; uninstall: `unins000.exe /VERYSILENT`.
- No-install version: `Velivo-1.22.exe` (a single file) or `Velivo-1.22-portable.zip`.
- **The files are not digitally signed** – Windows may show "Unknown publisher" → "More info" → "Run anyway". The release SHA-256 checksums are in `docs/VELIVO-SECURITY-AUDIT-CERTIFICATE.md`.

17. What Velivo doesn't have (as of 1.22)

- Faking fingerprinting readings (there is detection and a report).
- A "shared environment panel" for many computers (there is a LAN diagnostic panel).
- Syncing open tabs (there is "Send to…").
- A connection to the bank – payments are marked manually.
- A digital signature on the program files.

18. Video player and TV

18.1 Player

- A player based on libVLC (`VlcPlayer.cs`), installed optionally to `{app}\vlc`. Double-clicking a video file (associated with Velivo) opens just the player – the browser window stays minimised.
- A video library with resuming, custom titles and videos from another computer on the home network (`LanVideo.cs`).

18.2 TV

- Channels from the user's own list (`Iptv.cs`), TV guide, favourites, history of the last 20 channels, catch-up, recording, scheduling and a sleep timer. Access tokens are never written to logs.

18.3 Subtitles

- OpenSubtitles.com (`OpenSubtitles.cs`): search by file hash and by name, saved next to the film as `film.pl.srt`; the user's own account, password encrypted with DPAPI (`napisy-konto.bin`).
- Subtitle shift G / H remembered per film; subtitles can be hidden on the picture.

18.4 Narrator

- `SubtitleLector.cs`: reads subtitles with a Piper voice (offline) or a Windows voice, lowers the film's volume while reading; settings remembered per film.

19. On-screen keyboard and sticky notes

19.1 Keyboard

- `ScreenKeyboard.cs`: a non-activating window (doesn't take the cursor), characters injected with `SendInput`; keyboard listening (`WH_KEYBOARD_LL`) only while the keyboard is open or hidden, on its own thread; corrections run synchronously at the moment Space is pressed.
- Opening: the wheel, holding the wheel, the tray icon, the `Velivo.exe --klawiatura` shortcut ("Velivo Klawiatura" in the Start menu, its own AppUserModelID), optionally a click in a text field (UI Automation).
- Password fields are detected every 0.3 s (UI Automation) – then no corrections, learning or suggestions.

19.2 Dictionary, autocorrect and learning

- `SwipeTyping.cs`: Polish dictionary of about 90k words (only forms correct according to SJP), English; corrections: Polish letters, typos (edit distance 1), „nie”, fixed phrases, „ze/że” by context, polite forms of address, your own corrections.
- Learning (words, pairs and triples of words, sentences, corrections, My dictionary) stored in one DPAPI-encrypted file.

19.3 Gamepad

- XInput (`xinput1_4.dll`), read every 16 ms; right stick – cursor (`SetCursorPos`), clicks via `SendInput`; settings in preferences (`klawiatura-pad-*`).

19.4 Sticky notes

- `StickyNotes.cs`: saved in `karteczki.json`; due-date reminder – a date in the text (dd.mm[.yyyy], yyyy-mm-dd, dd/mm/yyyy), checked every 10 min, the "seen" state saved per day.